🧩 Detailed BRD: Transaction Tracking

Overview

The Transaction Tracking module serves as the primary auditing and accountability tool for the hardware store owner. Every time a sale is finalized in the POS module, a permanent record is generated here. This allows the Admin to review daily sales performance, investigate discrepancies, and verify exactly which staff member processed specific transactions.

Scope

- Automated logging of all finalized POS sales
- Displaying a chronological master list of historical transactions
- Viewing the itemized breakdown of a specific transaction (items sold, quantities, and unit prices)
- Filtering transaction history by date range and by specific cashier (Staff username)
- Calculating the total sales revenue for a filtered view

Out-of-Scope

- Processing returns, voids, or refunds directly through the transaction log
- Printing physical paper receipts from historical records
- Direct API integration with third-party accounting software
- Tracking customer names or buyer profiles associated with the transaction

Functional Requirements

Requirement	Actor	Priority	Rationale
The system shall automatically generate a unique Transaction ID for every finalized sale.	System	Must	Provides a clear reference number for auditing.

The system shall record the exact timestamp, total amount, and the active staff's username for every transaction.	System	Must	Enforces staff accountability and tracks peak sales hours.
The system shall display a Transaction History dashboard listing all completed sales chronologically.	Admin	Must	Allows the store owner to review business performance at a glance.
The system shall allow the user to click on any transaction record to view its itemized contents (items, quantities, unit prices).	Admin	Should	Essential for investigating specific sales or verifying inventory deductions.
The system shall allow the user to filter the transaction list by a specific date or date range.	Admin	Must	Allows the owner to calculate daily, weekly, or monthly revenue.

The system shall allow the user to filter the transaction list by a specific cashier (Staff username).	Admin	Must	Allows the Admin to audit the performance or investigate the shift of a specific employee.
The system shall display the total revenue of the currently filtered transactions on the dashboard.	System	Should	Provides an immediate financial summary without manual calculation.
The system shall completely hide the Transaction History dashboard from Staff-level accounts.	System	Should	Staff should only process current sales, not view overall store revenue data.

Non-Functional Requirements

Category	Requirement
Security	All transaction records cannot be edited, modified, or deleted by any staff.
Performance	Filtering a list of up to 100 transactions must complete and render within 1 second.

Data Retention	Transaction records must remain permanently linked to the Staff's username, even if that Staff account is later deactivated.
Usability	Date range filters shall default to the current day ("Today") upon initially loading the dashboard.

Constraints

- Because records are strictly immutable, cashiers who make a mistake during checkout cannot "void" the transaction in the system afterward; the Admin must issue a manual inventory adjustment to correct the stock.

Limitations

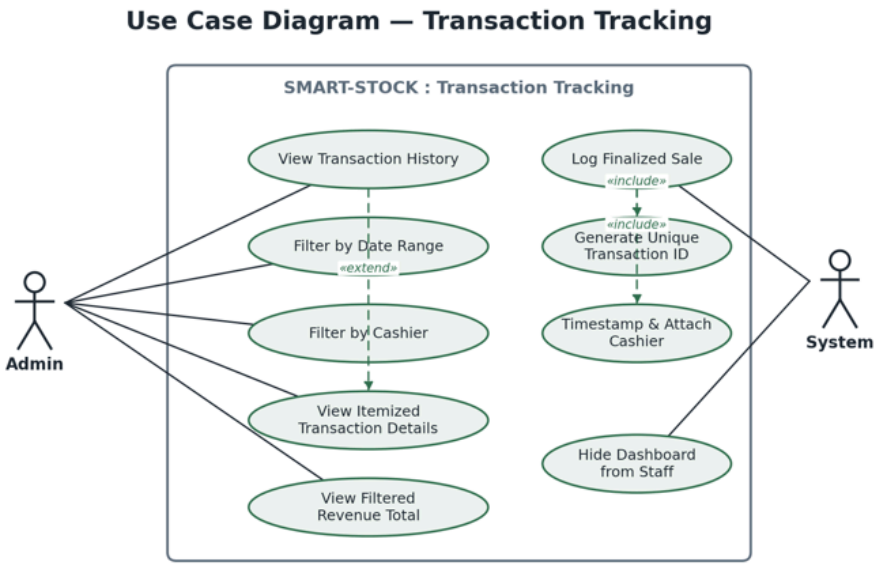
- The system relies entirely on the server's internal clock for timestamps; if the server time drifts, transaction logs will reflect the inaccurate time.

Assumptions

- Hardware store staff do not need to review their own past transactions once the current customer is cleared.
- Revenue tracking is based purely on the gross total of items sold; tax calculations and net profit margins (deducting wholesale cost) are handled externally by the owner.

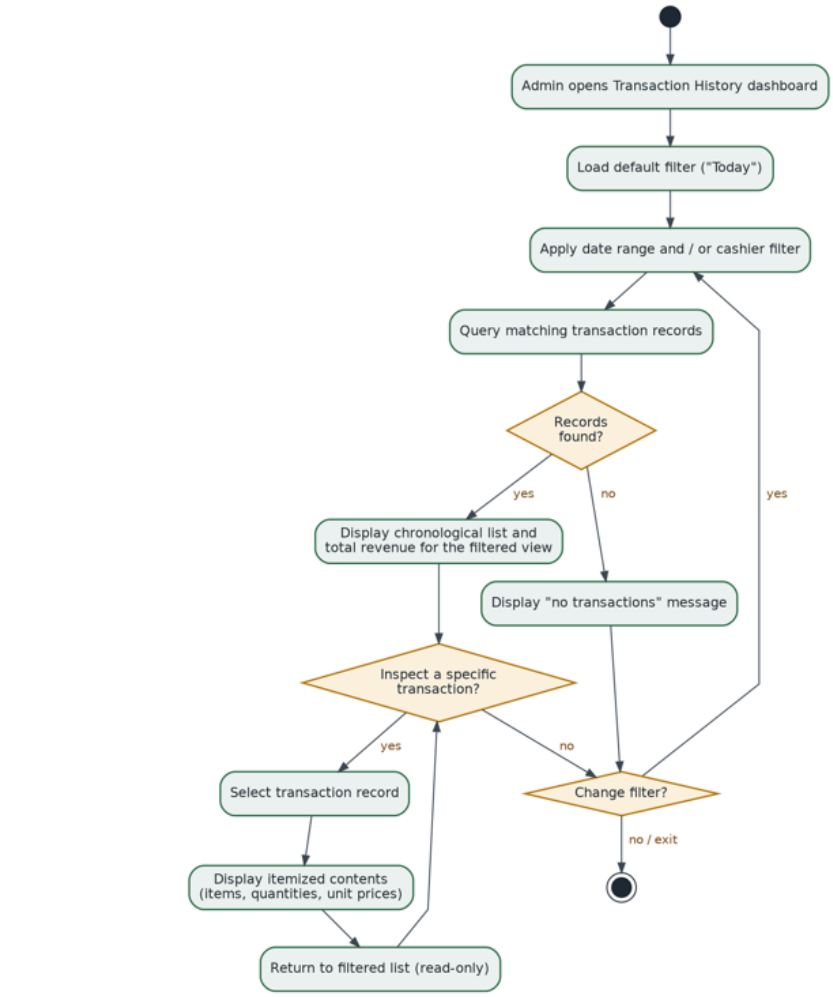
Requirements Model

- **Use Case Diagram**



- **Business rules**
  - Transaction Record Status = Read-Only.
  - Revenue Total = Sum of all Transaction Grand Totals within the applied filter parameters.
- **Activity Diagram**
  - *Transaction Auditing Workflow*

Activity Diagram — Transaction Auditing Workflow (Transaction Tracking)



- **State model**
  - *Transaction Log Lifecycle*

State Model — Transaction Log Lifecycle (Transaction Tracking)

